Proposal: Counterfactual Visual Analytics for Targeted Student Interventions

CSCE 679 Data Visualization

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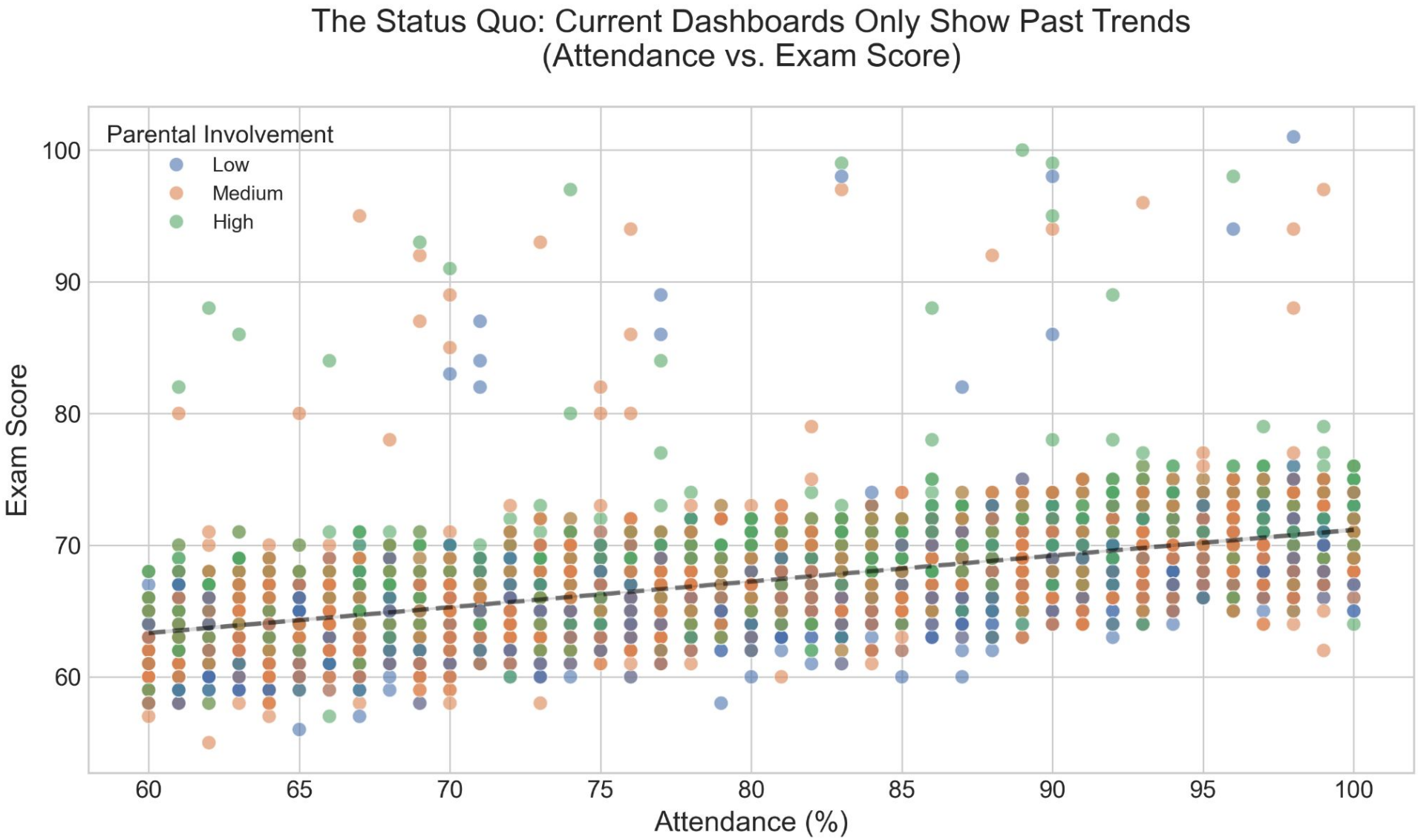
The problems with current dashboards



Why Existing Dashboards Fall Short

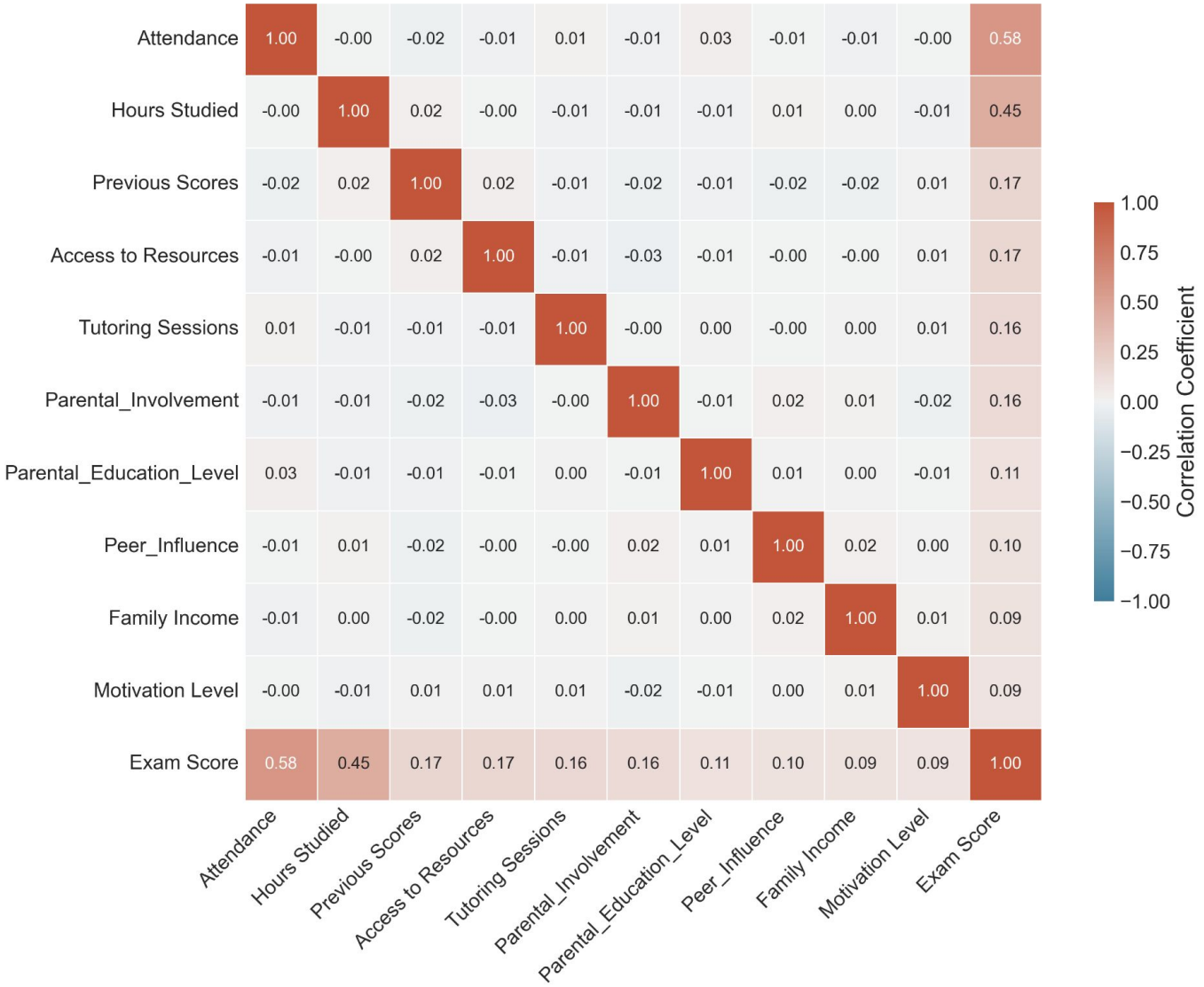
- Canvas: show grades, attendance, averages
- All descriptive; they look backward, not forward
- A Prof sees a failing student… but then what?
- No guidance on which action will actually help

Dataset

Student Academic Performance Dataset by Ayesha Siddiqa on Kaggle

Feature Category	Variables (Examples)	Data Type	Counterfactual Actionability
Academic History	Previous_Scores, Exam_Score (Target)	Numeric	**Immutable** (Baseline Metrics)
Student Behavior	Hours_Studied, Attendance	Numeric	**Highly Mutable** (Direct Intervention)
Lifestyle & Habits	Sleep_Hours, Extracurricular_Activities	Numeric / Cat.	**Mutable** (Counseling Focus)
Environment & Support	Tutoring_Sessions, Parental_Involvement	Categorical	**Partially Mutable** (Resource Focus)

The complexity of student success



Student Success is Highly Multivariate

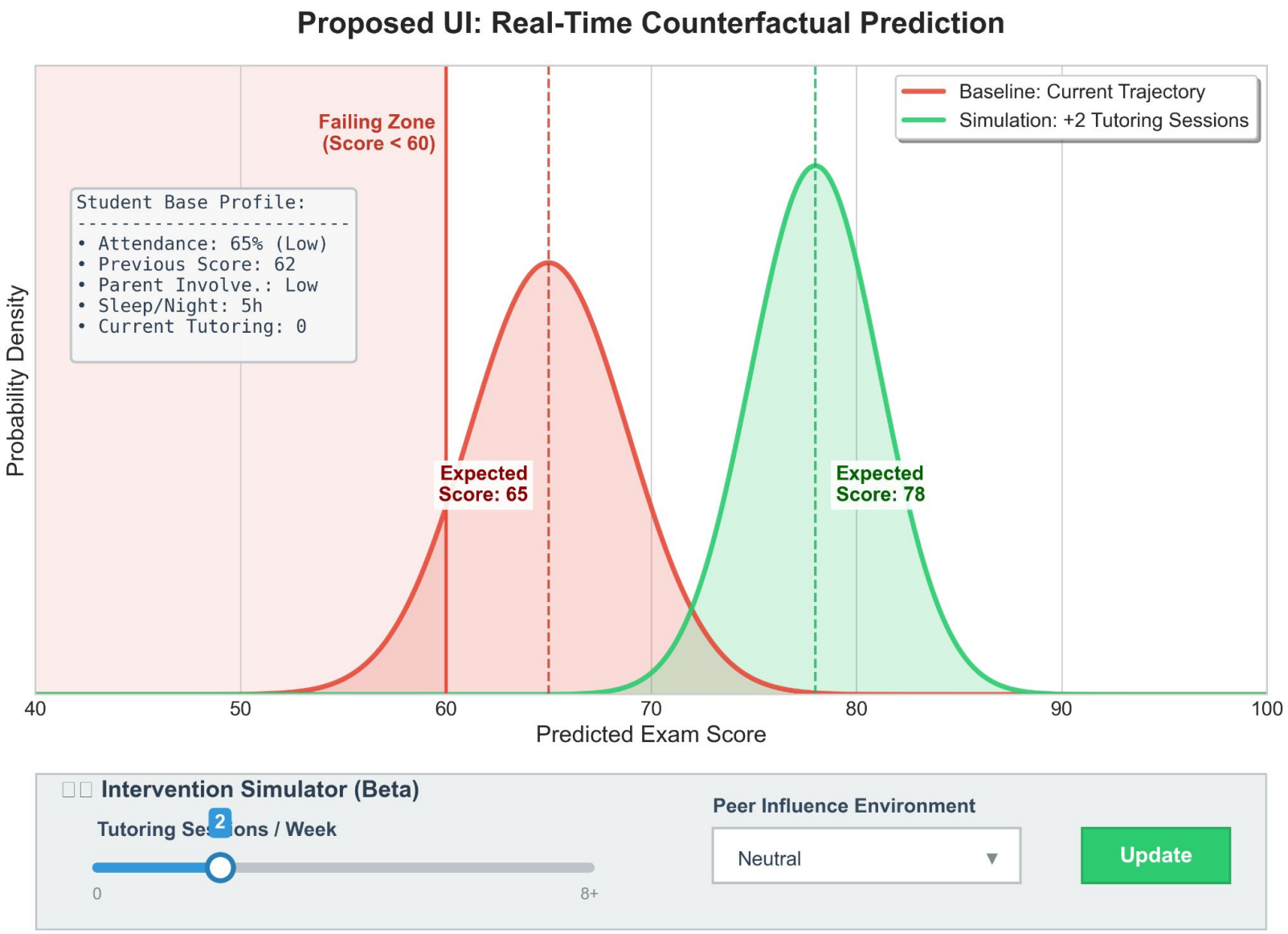
- 20 factors recorded per student
- Behavioral: Hours Studied, Attendance, Sleep Hours, Tutoring Sessions
- Structural: Parental Involvement, Family Income, School Type
- No single factor explains the full picture
- A regression model is needed to weigh them together

Proposed solution

Causal-Predictive Visual Analytics:

- Multiple linear regression model with bootstrapped confidence interval
- Interactive sliders for actionable variables (Tutoring, Attendance…)
- Real-time update of predicted score distribution
- Transparent: each coefficient readable as "+X pts per tutoring session"

The proposed UI



The counterfactual "what-if" visualization

Target users & design requirements

Target users:

- Professors, academic advisors, class instructors

How to gather requirements:

- Semi-structured interviews with 2-3 professors/instructors
- Show low-fidelity wireframes of the interface
- Ask: "Walk me through how you advise a struggling student today"
- Identify which variables they actually use and then map to dashboard design

Evaluation and timeline

Evaluation

- The Setup: Compare Baseline (spreadsheet) vs. Our Innovation (interactive dashboard)
- The Task: "You have 5 minutes. Formulate an intervention plan for 3 at-risk students."
- Quantitative Metrics:
    - Time-to-Decision (measured via stopwatch)
    - Confidence Level (1-10 self-reported scale)

Timeline

Weeks	Task
1–2	User interviews + data preprocessing + regression model fitting
3–4	Frontend dashboard (D3.js)
5–6	Integrate regression backend + bootstrapping + slider interactions
7–8	User evaluation + final report

Expected Results

Target: Increased Advisor Confidence

We anticipate that integrating counterfactual visual analytics will fundamentally shift how advisors interact with predictive data, empowering decision makers.

Expected Outcomes:

- Advisors will formulate intervention plans significantly faster and with higher confidence.
- The transparent nature of counterfactuals will increase overall trust in institutional AI models.
- Ultimately, providing realistic, data-backed roadmaps will positively impact retention rates.